Jeongwon Her

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Summary

ML engineer who takes fine-tuned open models to production: serves LoRA-adapted and quantized LLM/VLMs with vLLM and TensorRT-LLM (Multi-LoRA, W4A16, W8A16) on a design platform with 1M+ MAU. Background in quantization research (QAT, hardware-aware NAS; DAC 2026) and Kubernetes GPU infrastructure for distributed training.

Experience

MiriCanvas (MIRIDIH) | *ML Engineer, Inference Optimization* Aug 2025 – Present

Canva-like all-in-one design platform with 1M+ MAU and 20M+ registered users

- Took the team's LoRA-fine-tuned design-generation adapters (Qwen3-VL-32B) into a production batch-inference service with vLLM Multi-LoRA, and built its control plane for on-demand A100x8 instances across cloud regions: VPC integration, nginx load balancing, availability checks, and capacity planning (10K–60K requests/day per instance).
- Fit Qwen3.8-27B chat serving on a single 24 GB A10G (W4A16 Marlin, vLLM): **27 tok/s per stream**, about 70% of the memory-bandwidth bound, via CUDA-graph tuning (3.4x TPOT over eager); FP8 KV cache gave 2.5x capacity for 64K context.
- Patched TensorRT-LLM 1.3 RC to support Qwen3-Embedding-8B and served it in W8A16 with dynamic token lengths: latency from 100 ms (BF16, padded to 256 tokens) to **15 ms (6.7x)** on A10G for ∼30-token search queries at **100+ QPS**, about 80% of the memory-bandwidth bound.

Seoul National University | *Graduate Researcher, Inference Optimization* Sep 2023 – Aug 2025

- Recovered INT8 accuracy on accelerators via QAT, knowledge distillation, and tensor-scale recalibration.
- Built hardware-aware search: the genetic-algorithm NAS (DAC 2026) and a mixed-precision/pipeline search for operator-to-device mapping across GPU and NPU.
- Wrote an $O(N)$ hash-table CUDA voxelization kernel: 0.196 ms for 200K points, **3.5x** faster than the best existing kernel and 303x faster than the deterministic $O(N^2)$ one, with identical output below voxel caps.
- Deployed 11 vision models with TensorRT (INT8/FP16) and built inference engines for NVIDIA Jetson and the Telechips N-Dolphin automotive NPU; open-sourced as Jetson-CV-Opt.

Upstage | *MLOps Engineer* Sep 2022 – Aug 2023

Korea's leading LLM startup and first generative-AI unicorn, developer of the Solar LLM

- Maintained a 512-GPU Kubernetes farm (A100, RTX 3090; 4 clusters of 16 nodes) with ArgoCD GitOps on the MLOps team.
- Ran distributed training with DeepSpeed (ZeRO, TP/PP) via Hugging Face on NVLink-connected A100 nodes.
- Built a Python model-serving API on Kubernetes, plus dev/test environments with minikube and Skaffold.
- Enabled S3-compatible access on the WEKA parallel storage cluster and set per-team bucket/IAM policies for data access.

SNUAILAB | *ML Engineer* Sep 2021 – Aug 2022

Seoul National University spin-off for vision AI and video surveillance

- Built a semi-supervised training pipeline (Airflow, MLflow) and FastAPI training/inference services.

Publications

- Jisup Lee, Soonhoi Ha, **Jeongwon Her**, *Hardware-Aware Neural Architecture Search for Real-Time BEV-Based 3D Object Detection*, 63rd Design Automation Conference (**DAC**), 2026.
- **Jeongwon Her**, et al., *3D Object Detection Using the N-Dolphin Embedded NPU*, Korea Computer Congress (KCC), 2024.

Teaching

Samsung AI Expert Program | *Teaching Assistant* Aug 2024 – Dec 2024

- Taught Samsung Electronics engineers pruning and quantization; advised on compression for Galaxy RF modules.

Education

Seoul National University | M.S. in Computer Science and Engineering 2025
Dongguk University | B.S. in Computer Science and Engineering 2021

Skills

Inference & Quantization: vLLM, TensorRT-LLM, TensorRT, Multi-LoRA, W4A16, W8A16, FP8 KV cache, QAT
Training & Infra: PyTorch, DeepSpeed, Hugging Face, Kubernetes, KServe, ArgoCD **Languages:** Python, C++, CUDA